English Spanish

4.3 Comparing and Ordering Fractions, Decimals, and Percents

STATE STANDARDS
MA.6.A.5.1
MA.6.A.5.2

Essential Question How can you order numbers that are written as fractions, decimals, and percents?

1 ACTIVITY: Ordering Numbers

Work with a partner to order the following numbers.

$\frac{1}{8}$ 11% $\frac{3}{20}$ 0.172 0.32 43% 7% 0.7 $\frac{5}{6}$

a. Decide on a strategy for ordering the numbers. Will you write them all as fractions, decimals, or percents?

b. Use your strategy and a number line to order the numbers from least to greatest. (Note: Label the number line appropriately.)



2 ACTIVITY: Using Fractions, Decimals, and Percents

Work with a partner. Decide which number form (fraction, decimal, or percent) is more common. Then find which is greater.



a. 7% sales tax or $\frac{1}{20}$ sales tax



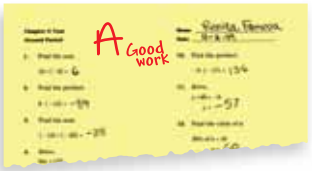
b. 0.37 cup of flour or $\frac{1}{3}$ cup of flour

c. $3\frac{5}{8}$-inch wrench or 3.69-inch wrench





d. $12\frac{2}{3}$ dollars or 12.56 dollars



e. 93% test score or $\frac{7}{8}$ test score



f. $5\frac{5}{6}$ fluid ounces or 5.6 fluid ounces

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3 ACTIVITY: The Game of Math Card War

Preparation:

- Cut index cards to make 40 playing cards.
- Write each number in the table on a card.

To Play:

- Play with a partner.
- Deal 20 cards to each player face-down.
- Each player turns one card face-up. The player with the greater number wins. The winner collects both cards and places them at the bottom of his or her cards.
- Suppose there is a tie. Each player lays three cards face-down, then a new card face-up. The player with the greater of these new cards wins. The winner collects all ten cards and places them at the bottom of his or her cards.
- Continue playing until one player has all the cards. This player wins the game.

75%	$\frac{3}{4}$	$\frac{1}{3}$	$\frac{3}{10}$	0.3	25%	0.4	0.25	100%	0.27
0.75	$66\frac{2}{3}\%$	12.5%	40%	$\frac{1}{4}$	4%	0.5%	0.04	$\frac{1}{100}$	$\frac{2}{3}$
0	30%	5%	$\frac{27}{100}$	0.05	$33\frac{1}{3}\%$	$\frac{2}{5}$	0.333. . .	27%	1%
1	0.01	$\frac{1}{20}$	$\frac{1}{8}$	0.125	$\frac{1}{25}$	$\frac{1}{200}$	0.005	0.666. . .	0%

What Is Your Answer?

4. **IN YOUR OWN WORDS** How can you order numbers that are written as fractions, decimals, and percents? Give an example with your answer.

5. All but one U.S. coin has a name that is related to its value. Which one is it? How are the names of the others related to their values?



Practice Use what you learned about ordering numbers to complete Exercises 4–7, 16, and 17 on page 164.

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4.3 Lesson

Check It Out Lesson Tutorials BigIdeasMath.com

When comparing and ordering fractions, decimals, and percents write the numbers as all fractions, all decimals, or all percents.

EXAMPLE 1 Comparing Fractions, Decimals, and Percents

a. Which is greater, $\frac{3}{20}$ or 16%?

Study Tip

It is usually easier to order decimals or percents.

Write $\frac{3}{20}$ as a percent: $\frac{3}{20} \overset{\times 5}{=} \frac{15}{100} = 15\%$

$\therefore$ 15% is less than 16%. So, 16% is the greater number.

b. Which is greater, 79% or 0.08?

Write 79% as a decimal: $79\% = 79.\% = 0.79$

$\therefore$ 0.79 is greater than 0.08. So, 79% is the greater number.

On Your Own

Now You're Ready
Exercises 4–15

1. Which is greater, 25% or $\frac{7}{25}$?
2. Which is greater, 0.49 or 94%?

EXAMPLE 2 Real-Life Application

You, your sister, and a friend each take the same number of shots at a soccer goal. You make 72% of your shots, your sister makes $\frac{19}{25}$ of her shots, and your friend makes 0.67 of his shots. Who made the fewest shots?

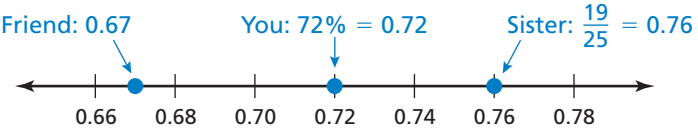
Remember

To order numbers from least to greatest, write them as they appear on a number line from left to right.

Write 72% and $\frac{19}{25}$ as decimals.

You: $72\% = 72.\% = 0.72$ **Sister:** $\frac{19}{25} \overset{\times 4}{=} \frac{76}{100} = 0.76$

Graph the decimals on a number line.



$\therefore$ 0.67 is the least number. So, your friend made the fewest shots.

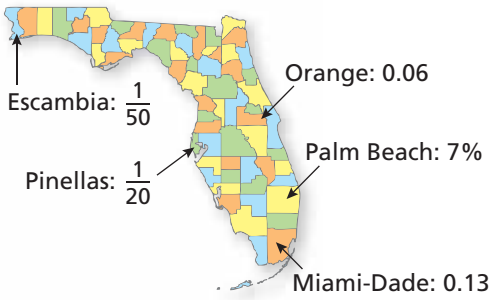
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On Your Own

Now You're Ready
Exercises 16–21

3. WHAT IF? You make 75% of your shots, your sister makes $\frac{13}{20}$ of her shots, and your friend makes 0.7 of his shots. Who made the most shots?

EXAMPLE 3 Real-Life Application



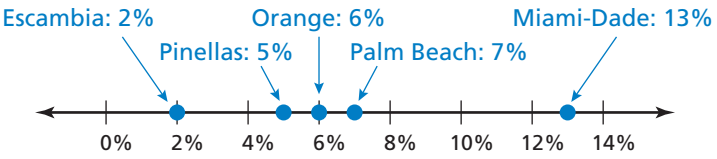
The map shows the portions of the population of Florida that live in five counties.

List the five counties in order by population from least to greatest.

Begin by writing each portion as a fraction, decimal, and percent.

County	Fraction	Decimal	Percent
Escambia	$\frac{1}{50}$	0.02	2%
Orange	$\frac{6}{100}$	0.06	6%
Pinellas	$\frac{1}{20}$	0.05	5%
Palm Beach	$\frac{7}{100}$	0.07	7%
Miami-Dade	$\frac{13}{100}$	0.13	13%

Graph the percent for each county on a number line.



∴ The counties in order from least to greatest are Escambia, Pinellas, Orange, Palm Beach, and Miami-Dade.

On Your Own

4. The portion of the population of Florida that lives in Duval County is 0.046. The portion that lives in Brevard County is $\frac{3}{100}$. Reorder the counties in Example 3 including Duval and Brevard counties.

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4.3 Exercises

Check It Out
Help with Homework
BigIdeasMath.com

Vocabulary and Concept Check

1. **NUMBER SENSE** Copy and complete the table.

Fraction	Decimal	Percent
$\frac{18}{25}$	0.72	
$\frac{17}{20}$		85%
$\frac{13}{50}$		
	0.62	
		45%

2. **NUMBER SENSE** How would you decide whether $\frac{3}{5}$ or 59% is greater? Explain.

3. **WHICH ONE DOESN'T BELONG?** Which one does *not* belong with the other three? Explain your reasoning.

40% $\frac{2}{5}$ 0.4 0.04

Practice and Problem Solving

Tell which number is greater.

4. 0.9, 95%
5. 20%, 0.02
6. $\frac{37}{50}$, 37%
7. 50%, $\frac{13}{25}$
8. 0.086, 86%
9. 76%, 0.67
10. 60%, $\frac{5}{8}$
11. 0.12, 1.2%
12. 17%, $\frac{4}{25}$
13. 140%, 0.14
14. $\frac{1}{3}$, 30%
15. 80%, $\frac{7}{9}$

~~Use a number line~~ to order the numbers from least to greatest.

16. 38%, $\frac{8}{25}$, 0.41
17. 68%, 0.63, $\frac{13}{20}$
18. $\frac{43}{50}$, 0.91, $\frac{7}{8}$, 84%
19. 0.15%, $\frac{3}{20}$, 0.015
20. 2.62, $2\frac{2}{5}$, 26.8%, 2.26, 271%
21. $\frac{87}{200}$, 0.44, 43.7%, $\frac{21}{50}$

22. **TEST** You answer 21 out of 25 questions correctly on a test. Did you reach your goal of getting 80% or better?

23. **POPULATION** The table shows the portions of the world population that live in four countries. Order the countries by population from least to greatest.

Country	Brazil	China	Japan	United States
Portion of World Population	3%	$\frac{1}{5}$	$\frac{1}{50}$	0.046

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Order the numbers from least to greatest.

24. $66.1\%,\ 0.66,\ \frac{2}{3},\ 0.667$

25. $\frac{2}{9},\ 21\%,\ 0.2\overline{1},\ \frac{11}{50}$

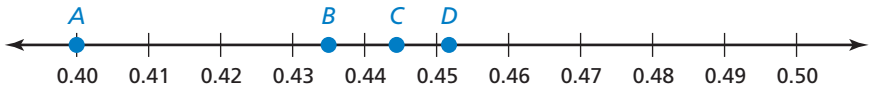
Tell which letter shows the graph of the number.

26. $\frac{2}{5}$

27. 45.2%

28. 0.435

29. $\frac{4}{9}$



30. TOUR DE FRANCE The Tour de France is a bicycle road race. The whole race is made up of 21 small races called stages. The table shows how several stages compare to the whole Tour de France. Order the stages from shortest to longest.

Stage	1	7	8	17	21
Portion of Total Distance	$\frac{11}{200}$	0.044	$\frac{6}{125}$	0.06	4%

31. SLEEP The table shows the portions of the day that several animals sleep.

a. Order the animals by sleep time from least to greatest.

b. Estimate the portion of the day that you sleep.

c. Where do you fit on the ordered list?



Animal	Portion of Day Sleeping
Dolphin	0.433
Lion	56.3%
Rabbit	$\frac{19}{40}$
Squirrel	$\frac{31}{50}$
Tiger	65.8%

32. Number Sense Tell what whole number you can substitute for a in each list so the numbers are ordered from least to greatest. If there is none, explain why.

a. $\frac{2}{a},\ \frac{a}{22},\ 33\%$

b. $\frac{1}{a},\ \frac{a}{8},\ 33\%$

Fair Game Review What you learned in previous grades & lessons

Multiply. SECTION 3.2

33. 0.6×8

34. 3.3×5

35. 0.74×9

36. 2.19×12

37. MULTIPLE CHOICE A flute is 27 inches long. A piccolo is $12\frac{1}{2}$ inches long. How many times greater is the length of the flute than the length of the piccolo? SECTION 2.6

(A) $\frac{25}{54}$ (B) $1\frac{1}{5}$ (C) $2\frac{4}{25}$ (D) 6

Ordering Fractions, Decimals and Percentages

Name:	Class:	Date:

Mark	/ 5	%

1) Order this list from smallest to largest. [1]

0.3 15% 2 $\frac{1}{2}$

2) Order this list from smallest to largest. [1]

$\frac{3}{4}$ 3 0.4 5%

3) Order this list from smallest to largest. [1]

0.14 4% 0.48 $\frac{1}{5}$

4) Order this list from smallest to largest. [1]

0.46 3% $\frac{1}{4}$ 0.14

5) Order this list from smallest to largest. [1]

$\frac{6}{5}$ 58.9% 1.77 0.169

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Solutions for the assessment Ordering Fractions, Decimals and Percentages

1) 15%, 0.3, $\frac{1}{2}$, 2

2) 5%, 0.4, $\frac{3}{4}$, 3

3) 4%, 0.14, $\frac{1}{5}$, 0.48

4) 3%, 0.14, $\frac{1}{4}$, 0.46

5) 0.169, 58.9%, $\frac{6}{5}$, 1.77

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Ratio, Rate and Proportions Practice

- **Ratios**

A ratio is a comparison of two quantities that have the same units. You can express a ratio in any one of the following ways:

$\frac{18}{5}$ 18:5 18 to 5

Example #1: If one store has 360 items and another store has 100 of the same items, express the ratio of the items.

$\frac{360}{100}$ or 360:100 or 360 to 100

Ratios are usually written in lowest terms; therefore, the above example would reduce in this way:

$\frac{360}{100} \begin{matrix} \div 20 \\ \div 20 \end{matrix}$ (What is the largest number you can divide both values by?)

$= \frac{18}{5}$

Example #2: John earns $350 a week. His take-home pay, however, is $295. What is the ratio of his gross pay to his take-home pay?

$\frac{350}{295} = \frac{70}{59}$

- **Rates**

A rate is a comparison of two quantities that have different units. Rates are usually expressed in the fractional form.

Example #3: Fran paid $16 for her 12-month subscription to Better Homes and Gardens magazine. Express as a rate.

$\frac{\$16.00}{12 \text{ magazines}} = \frac{\$4.00}{3 \text{ magazines}}$

If Francine wants to know how much she pays for each (1) magazine, she can divide $4 by 3 magazines. This will give her the price per magazine (also called the unit rate).

$4.00 = $1.33/magazine

- **Proportions**

A proportion is a statement that two ratios or rates are equal. It can be given as a sentence in words, but most often a proportion is an algebraic equation.

The equation: $\frac{3}{5} = \frac{21}{35}$ is a proportion because its cross-products are equal.

$3 \times 35 = 105$ and $5 \times 21 = 105$

Proportions are solved by using this cross-product rule.

<u>Practice:</u> For each word problem, write and then solve the proportion to find the answer.

A. Write each ratio as a fraction in lowest terms.

1. 2 to 4
2. $\frac{15}{20}$
3. 6:18
4. 21:15
5. $\frac{12}{18}$
6. 3 to 12
7. 7: 4
8. $\frac{18}{12}$
9. 20:16
10. 15 to 36
11. 35:7
12. $\frac{8}{28}$
13. 24 to 96
14. 9:27
15. $\frac{11}{88}$

B. Write each of the following rates as a unit rate.

1. $\frac{3 \text{ Tbsp}}{2 \text{ tsp}}$
2. $\frac{135 \text{ pitches}}{45 \text{ strikes}}$
3. $\frac{128 \text{ miles}}{4 \text{ hours}}$
4. $\frac{2250 \text{ pencils}}{18 \text{ boxes}}$
5. $\frac{\$450}{18 \text{ shares}}$
6. $\frac{2500 \text{ meters}}{15 \text{ seconds}}$
7. $\frac{\$5{,}082}{475 \text{ sq.yds.}}$
8. $\frac{750 \text{ gallons}}{14 \text{ minutes}}$

C. Solve each proportion and give the answer in simplest form.

1. $6 : 8 = n : 12$

2. $\frac{2}{7} = \frac{8}{n}$

3. $\frac{n}{6} = \frac{11}{3}$

4. $4 : n = 6 : 9$

5. $\frac{3}{n} = \frac{2}{5}$

6. $\frac{0.4}{1.5} = \frac{12}{n}$

7. $2\frac{1}{2} : 3\frac{1}{2} = n : 2$

8. $1 : 2 = n : 9$

9. $4 \text{ to } 8 = 15 \text{ to } n$

10. $18 : n = 3 : 11$

11. $\frac{5}{6} = \frac{n}{30}$

12. $\frac{12}{40} = \frac{n}{25}$

13. $8 : 19 = 14 : n$

14. $\frac{10}{n} = \frac{2}{1.7}$

15. $24 : \frac{1}{4} = n : \frac{1}{3}$

16. $44 \text{ to } 121 = n \text{ to } 11$

D. Solve by using a proportion. Round answers to the nearest hundredth if necessary.

1. You jog 3.6 miles in 30 minutes. At that rate, how long will it take you to jog 4.8 miles?

2. You earn $33 in 8 hours. At that rate, how much would you earn in 5 hours?

3. An airplane flies 105 miles in ½ hour. How far can it fly in 1 ¼ hours at the same rate of speed?

4. What is the cost of six filters if eight filters cost $39.92?

5. If one gallon of paint covers 825 sq. ft., how much paint is needed to cover 2640 sq. ft.?

6. A map scale designates 1” = 50 miles. If the distance between two towns on the map is 2.75 inches, how many miles must you drive to go from the first town to the second?

7. Bob is taking his son to look at colleges. The first college they plan to visit is 150 miles from their home. In the first hour they drive at a rate of 60 mph. If they want to reach their destination in 2 ½ hours, what speed must they average for the remainder of their trip?

8. Four employees can wash 20 service vehicles in 5 hours. How long would it take 5 employees to wash the same number of vehicles?

9. These two figures are similar. Use a proportion to find the length of side *n.*



Ratio

a relationship btw 2 #'s.

3 forms

$\frac{A}{B}$ A:B A to B

Keep the same order as the words

What is the ratio of little boys to girls in this room?

$\frac{9}{10}$ 9:10 9 to 10

Girls to boys

10:9

Ratio of girls to human beings

$\frac{10}{20} = \frac{1}{2}$

10 to 20

1 to 2

Rate

is a ratio

w labels

Speed

$$\frac{55 \text{ Miles}}{1 \text{ Hour}}$$

$$\frac{36 \text{ laps}}{6 \text{ Mintes}}$$

Name_____ Date _____

Measuring Rate

Key Ideas

Fill in the blanks with the term or terms that makes each statement true.

A _____ is a comparison of the size of one number as compared to the size of another number.

Two ratios are said to be _____ if they have the same value.

A ratio that compares two different units is called a(n) _____.

A rate that compares a quantity per *one* of the other quantity is called a(n) _____.

Word Bank
unit rate
ratio
rate
equivalent

Review

Determine which equations show a true statement. If the fractions are equivalent, put a "✓" next to them. If they are not equivalent, put an "X" through them.

a. $\frac{1}{2} = \frac{6}{12}$ b. $\frac{4}{5} = \frac{12}{20}$ c. $\frac{4}{18} = \frac{2}{9}$ d. $\frac{6}{10} = \frac{15}{25}$

Guided Practice

Follow the example to find the unit rate for each rate shown below.

Rate:
a ratio comparing two different units

division sign

12 miles / 3 hours

we read this as "12 miles *per* 3 hours"

Unit Rate:
the first quantity per *one* of the second quantity

12 ÷ 3 = 4

4 miles / 1 hour or 4 miles/hour

a. 9 gallons / 3 seconds

b. $40.00 / 4 hours

c. 40 heartbeats / 30 seconds

d. 100 steps / 2 minutes

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Practice

1. Write down the rate in each scenario. Then, find the unit rate.

a. buying 12 pencils for $3.00

b. traveling 80 miles in 2 hours

c. walking 2 miles in 24 minutes

d. peeling 8 potatoes in 5 minutes

2. Are the rates and unit rates in each scenario above considered to be equivalent? Why or why not?

3. Determine which pairs of ratios below are equivalent. If the ratios are equivalent, put a "✓" next to them. If they are not equivalent, put an "X" through them.

a. 5 to 3, 20 to 12

b. $\frac{14}{22}$, $\frac{7}{11}$

c. 6:4, 24:8

4. For each ratio below, write an equivalent ratio.

a. 18 to 30

b. 50:200

c. $\frac{2}{3}$

d. 60 to 12

e. $\frac{1}{4}$

f. 70:30

g. 32:8

h. 350 to 500

i. $\frac{40}{10}$

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5. Seth bakes 100 cookies in 2 hours. Erica bakes 200 cookies in 2.5 hours.

a. Write the rate at which each person is baking cookies.

b. Write the unit rate at which each person is baking cookies.

c. Which person is baking at a faster rate? How you know?

6. Brandon can swim 15 laps around the pool in 40 minutes. Peter can swim 10 laps in 25 minutes.

a. Write the rate at which each boy swims.

b. Write the unit rate at which each boy swims.

c. If both boys continue swimming, which boy will swim 20 laps first? How do you know?

7. Jasmine buys a pack of 15 socks for $9.00. Shawn buys a pack of 20 socks for $11.00. Write a ratio to represent they paid to the number of socks. Who got a better deal on socks? Explain.

8. Mario and his 3 friends attend a professional football game. After parking, tickets, and food, they spent $240.00 total. Mario tells his friends that he got a better deal last week, paying only $50/person. How much better of a deal did Mario get last week?

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DISTANCE, SPEED AND TIME – PRACTICE QUESTIONS

1.
A car travelled 70 miles in 2 hours.
Work out the car's average speed, in miles per hour.

2.
A train travelled from Bristol to London.
The distance from Bristol to London is 120 miles.
The train travelled at an average speed of 40 miles per hour.
How long did the journey take?

3.
A long distance runner ran at an average speed of 10 miles per hour for two and a half hours.
How many miles did the runner run?

4.
A sprinter ran 200 metres in 40 seconds.
Work out the sprinter's average speed, in metres per second.

5.
Scott drove at an average speed of 35 miles per hour for 3 hours.
How many miles did Scott drive?

6.
Tori cycled 9 kilometres in 2 hours.
Work out Tori's average speed, in kilometres per hour.

7.
A plane travelled at an average speed of 400 miles per hour for three and a half hours.
How many miles did the plane travel?

8.
A car travelled 225 kilometres at an average speed of 45 kilometres per hour.
How long did the journey take?

9.
A helicopter travelled at an average speed of 110 miles per hour for an hour and a half.
How many miles did the helicopter travel?

10.
A boat travelled 34 miles in 8 hours.
What was the boat's average speed?

11.
Amy drove at an average speed of 45 miles per hour for 20 minutes.
How many miles did she drive?

12.
Beth cycled at an average speed of 12 kilometres per hour for 90 minutes.
How many kilometres did she cycle?

13.
Craig drove 90,000 metres at an average speed of 45 kilometres per hour.
How many hours was he driving for?

14.
David ran 8 kilometres in 50 minutes.
Work out David's average speed, in metres per second.

15.
Evelyn drove 60 kilometres at an average speed of 500 metres per minute.
How many hours was she driving for?

16.
Farah ran at an average speed of 2.8 metres per second for 30 minutes.
How many metres did Farah run?

17.
Greg drove 18 kilometres in 20 minutes.
Work out Greg's average speed, in metres per second.

18.
Henrietta drove 3,600 metres in 8 minutes.
Work out Henrietta's average speed, in kilometres per hour.

19.
Kevin needs to get to work by 9am.
Kevin's workplace is 3 miles away from his house.
He is planning to cycle to work.
He can cycle at a maximum speed of 9 miles per hour.
Kevin says "If I leave my house at 8:45am and cycle as fast as I can, I will arrive on time".
Is Kevin right? Explain your answer.

20.
Esther and Freddie both cycled to school.
Esther lives 1,800 metres from the school.
Freddie lives 3 kilometres from the school.
Esther travelled at an average speed of 4 metres per second, and Freddie travelled at an average speed of 8 metres per second.
They both left their houses at the same time.
Who arrived at school first?

21.
Maria walks to work.
She walks at an average speed of 5 kilometres per hour.
She lives 3 kilometres away from her workplace.
She needs to be in work by 9am.
What is the latest time she can leave her house?

22.
Neil and Martin both drove from Birmingham to Manchester.
Neil left at 1:50 and Martin left at 1:20.
Neil drove at an average speed of 30 miles per hour and Martin drove at an average speed of 25 miles per hour.
The distance from Birmingham to Manchester is 45 miles.
Who arrived in Manchester first?

23.
Louise and Vanessa both drove the exact same route.
Louise drove at an average speed of 40 miles per hour for 150 minutes.
Vanessa drove at an average speed of 25 miles per hour.
How long did Vanessa drive for?

Comprehensive Mixed Conversions Master Set

10-Problem Real-World Applications: Fractions, Decimals, & Percents

Worksheet #105
10 Problem Practice

Student Name: ---------- **Date:** ---------- **Score:** ----------

Instructions: Solve each 10-problem scenario. Convert values across fractions, decimals, and percents as needed, but provide each final response strictly in the **REQUIRED FORMAT**.

PROBLEM 1: AEROSPACE MANUFACTURING — **Format: SIMPLIFIED FRACTION**

1. Aircraft Alloy Composition

An experimental titanium alloy designed for space exploration was synthesized using four primary metal components:

Component Metal	Element Symbol	Mass Share
Titanium	Ti	0.55
Aluminum	Al	3/10
Vanadium	V	12.5%
Iron (Trace)	Fe	*Remaining share*

Calculate the share of Iron (Fe) in the alloy. Express your answer as a **simplified fraction**.

Final Answer (Simplified Fraction): 1/40

PROBLEM 2: MARINE BIOLOGY — **Format: PERCENTAGE (%)**

2. Coral Reef Species Distribution

Marine researchers conducted a biodiversity survey across a 500-square-meter section of a protected coral reef system:

REEF SPECIES PROPORTION

7/20	0.275	20%	?

Hard Corals (7/20) · Soft Corals (0.275) · Sea Anemones (20%) · Sponges (Rest)

What **percentage** of the surveyed reef area is populated by **Sponges**?

Final Answer (Percentage %):

PROBLEM 3: SOFTWARE ENGINEERING — **Format: DECIMAL**

3. Mobile App Storage Usage

A smartphone operating system evaluated the memory consumption of a newly installed social media application:

- **User Data:** 0.38 of total space
- **Cached Media:** 2/5 of total space
- **Executable Code:** 14% of total space
- **Logs & Temp:** Remaining space

Calculate the share of storage taken up by **Logs & Temp files**. Express as a **decimal**.

Final Answer (Decimal):

PROBLEM 4: RENEWABLE ENERGY Format: WHOLE NUMBER (KILOWATT-HOURS)

4. Solar Microgrid Power Output

A community microgrid generated **3,600 Kilowatt-hours (kWh)** of clean solar energy during a summer day:

Sector Destination	Allocation Share	Format Type
Residential Neighborhoods	9/16	Fractional
Commercial Buildings	0.2875	Decimal
Municipal Streetlights	8.75%	Percentage
Battery Reserve Storage	*Remainder*	Unknown Share

.5625
.2875
+ .0875
.9375

Determine the exact energy sent to **Battery Reserve Storage** in whole **Kilowatt-hours (kWh)**.

1 − .9375 = .0625

6.25%

Final Answer (Whole kWh):

PROBLEM 5: FINANCIAL INVESTMENTS Format: DOLLARS & CENTS ($)

5. Portfolio Asset Allocation

An investment manager oversees an institutional portfolio total value of **$240,000** divided across four asset classes:

- **Index Funds:** 0.475 of total portfolio
- **Corporate Bonds:** 1/3 of total portfolio
- **Real Estate Investment Trusts (REITs):** 12.5% of total portfolio
- **Cash Reserves:** The remaining balance

Calculate the exact dollar value held in **Cash Reserves**.

Final Answer (Dollar Amount):

PROBLEM 6: AGRICULTURE & FARMING Format: SIMPLIFIED FRACTION

6. Agricultural Land Utilization

A 1,200-acre commercial farm partitioned its acreage for seasonal crop cultivation:

CROP HARVEST BREAKDOWN

- 🌾 **Corn:** 35% of total acreage
- 🌱 **Soybeans:** 0.325 of total acreage
- 🌾 **Wheat:** 1/5 of total acreage
- 🌻 **Sunflowers:** Remaining acreage

What proportion of the total farm acreage is dedicated to growing **Sunflowers**? Express as a **simplified fraction**.

Final Answer (Simplified Fraction):

PROBLEM 7: TRANSPORTATION & TRANSIT

Format: PERCENTAGE (%)

7. Commuter Mode Split

A metropolitan transit authority surveyed 10,000 daily commuters regarding their primary transport mode:

- **Subway / Rail:** 0.46 of commuters
- **Bus Transit:** 27.5% of commuters
- **Personal Bicycle / Walking:** 7/40 of commuters
- **Rideshare / Personal Vehicles:** The remaining commuters

What **percentage** of commuters use **Rideshare / Personal Vehicles**?

Final Answer (Percentage %):

PROBLEM 8: PUBLIC HEALTH

Format: DECIMAL

8. Nutritional Caloric Breakdown

A nutritionist evaluated the daily macronutrient energy distribution for an athlete consuming 2,800 total calories:

- **Complex Carbohydrates:** 11/20 of total calories
- **Lean Proteins:** 27.5% of total calories
- **Healthy Fats:** 0.14 of total calories
- **Dietary Fiber & Micronutrients:** Remaining calories

Find the share of total calories supplied by **Dietary Fiber & Micronutrients**. Express as a **decimal**.

Final Answer (Decimal):

PROBLEM 9: RETAIL & COMMERCE

Format: WHOLE NUMBER (UNITS)

9. Warehouse Inventory Audit

A fulfillment center holds **4,800 units** of electronics in stock prior to a major holiday sale event:

- **Smartphones:** 3/8 of total inventory
- **Tablets:** 0.2875 of total inventory
- **Laptops:** 22.5% of total inventory
- **Smartwatches:** Remaining inventory

How many individual **Smartwatches** are currently in stock?

Final Answer (Whole Number Units):

PROBLEM 10: ENVIRONMENTAL SCIENCE

Format: DOLLARS & CENTS ($)

10. Ecosystem Grant Distribution

An environmental defense fund awarded **$500,000** in research grants across four conservation projects:

- **Reforestation Initiative:** 0.385 of total grant funding
- **Ocean Debris Cleanup:** 31% of total grant funding
- **Wetland Restoration:** 9/40 of total grant funding
- **Wildlife Sanctuary Expansion:** Remaining funds

Determine the exact funding awarded to the **Wildlife Sanctuary Expansion** project.

Final Answer (Dollar Amount):

Comprehensive Solution Key (Problems 1–10)

Step-by-Step Derivations and Verification

1. Aircraft Alloy (Target: Simplified Fraction)

Ti (0.55) + Al (3/10 = 0.30) + V (12.5% = 0.125) = 0.975. Fe share = 1 - 0.975 = 0.025 = 25/1000 = **1/40**.

2. Coral Reef Species (Target: Percentage %)

Hard Coral (7/20 = 35%) + Soft Coral (0.275 = 27.5%) + Anemones (20%) = 82.5%. Sponges = 100% - 82.5% = **17.5%**.

3. Mobile App Storage (Target: Decimal)

User Data (0.38) + Cache (2/5 = 0.40) + Code (14% = 0.14) = 0.92. Logs & Temp = 1.00 - 0.92 = **0.08**.

4. Solar Microgrid Output (Target: Whole kWh)

Residential (9/16 = 0.5625) + Commercial (0.2875) + Lights (8.75% = 0.0875) = 0.9375. Reserve = 1 - 0.9375 = 0.0625. 3,600 × 0.0625 = **225 kWh**.

5. Portfolio Allocation (Target: Dollar Amount $)

Index (0.475 = 47.5%) + Bonds (1/3 = 33.333%) + REITs (12.5%) = 93.333% (or 112/120 = 14/15). Cash share = 1 - (19/40 + 1/3 + 1/8) = 1 - 112/120 = 8/120 = 1/15. $240,000 × (1/15) = **$16,000**.

6. Agricultural Land (Target: Simplified Fraction)

Corn (35% = 0.35) + Soybeans (0.325) + Wheat (1/5 = 0.20) = 0.875. Sunflowers = 1 - 0.875 = 0.125 = 125/1000 = **1/8**.

7. Commuter Mode Split (Target: Percentage %)

Rail (0.46 = 46%) + Bus (27.5%) + Bike/Walk (7/40 = 17.5%) = 91%. Rideshare = 100% - 91% = **9%**.

8. Nutritional Caloric Breakdown (Target: Decimal)

Carbs (11/20 = 0.55) + Protein (27.5% = 0.275) + Fats (0.14) = 0.965. Micronutrients = 1.000 - 0.965 = **0.035**.

9. Warehouse Inventory (Target: Whole Units)

Smartphones (3/8 = 0.375) + Tablets (0.2875) + Laptops (22.5% = 0.225) = 0.8875. Smartwatches = 1 - 0.8875 = 0.1125. 4,800 × 0.1125 = **540 units**.

10. Ecosystem Grant Distribution (Target: Dollar Amount $)

Reforestation (0.385 = 38.5%) + Cleanup (31%) + Wetland (9/40 = 22.5%) = 92%. Sanctuary = 100% - 92% = 8% (0.08). $500,000 × 0.08 = **$40,000**.